A Project Report

Submitted in Fulfilment of the Data Analytics Training Programme

E-Commerce Customer Behaviour Dashboard

Using Excel | Python | Power BI

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Course : Data Analytics

Batch : Batch 2

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1. Introduction

E-commerce has transformed global retail, generating enormous volumes of transactional data that, when properly analysed, can yield powerful insights into customer behaviour, purchasing patterns, and key revenue drivers. This project presents a comprehensive data analytics solution built on the UCI Machine Learning Repository's Online Retail dataset, a real-world dataset comprising over 540,456 transactions recorded between December 2010 and December 2011 for a UK-based online retailer.

The project follows a structured end-to-end analytics pipeline. Initially, raw data was cleaned, preprocessed, and structured using Microsoft Excel to ensure accuracy and consistency. This was followed by exploratory data analysis (EDA) in Python using Jupyter Notebook, where statistical summaries and visualizations were used to uncover trends, patterns, and anomalies in the dataset. Key analyses included product performance, customer purchasing behavior, country-wise revenue distribution, and time-based revenue trends.

Subsequently, the refined dataset was imported into Microsoft Power BI, where advanced data transformation techniques and DAX calculations were applied to develop an interactive and dynamic dashboard. The dashboard provides a clear visualization of key performance indicators (KPIs), including total revenue, customer segmentation, product performance, and geographic insights.

The result is a visually intuitive and business-ready analytical tool that enables efficient monitoring of sales performance and supports data-driven decision-making. This report documents the complete methodology, tools and technologies used, technical implementation, insights generated, and the overall business impact of the project, highlighting the importance of data analytics in enhancing strategic decision-making and improving business outcomes

2. Problem Statement

Online retail businesses accumulate vast amounts of raw transactional data that is often underutilized and not effectively transformed into meaningful, decision-support information. Without structured analysis and proper visualization, critical business questions remain unanswered:

- Which products generate the most revenue, and which should be promoted or discontinued?
- Which markets and geographies drive the majority of sales?
- What are the seasonal purchasing trends that should inform marketing strategies and campaign planning?
- Which customers are high-value, at risk of churn, or already lost?
- Is the business performing in line with its revenue targets?

In the absence of data-driven insights, organizations risk making uninformed decisions, leading to inefficient allocation of resources, missed revenue opportunities, and reduced customer retention. Additionally, the lack of clear visualization tools makes it difficult to interpret complex datasets and identify actionable patterns.

Therefore, there is a strong need for a structured analytics solution that can process large volumes of transactional data and present insights in a clear, interactive, and meaningful format. This project addresses this gap by developing a comprehensive data analytics and visualization system that transforms raw data into actionable business intelligence, enabling better decision-making and improved overall performance.

3. Objectives of the Project

To clean and prepare raw e-commerce transaction data to ensure accuracy and reliability for analysis.

To perform exploratory data analysis (EDA) in order to understand data distributions, trends, and potential anomalies.

To calculate and monitor key business KPIs, including Total Revenue, Total Orders, Unique Customers, Unique Products, and Countries served.

To segment customers into meaningful groups such as **Champions, Loyal, At Risk, and Lost**, based on their revenue contribution.

To identify top-performing products and major geographic contributors to overall revenue.

To analyze monthly and quarterly revenue trends in order to identify seasonal patterns and sales fluctuations.

To design and develop an interactive, filter-enabled dashboard using Power BI for effective data visualization.

To generate actionable insights that support strategic decision-making in marketing, sales, and business expansion

4. Scope of the Project

In Scope

- Transactional data from December 2010 to December 2011 (UCI Online Retail Dataset)
- Data cleaning, validation, and preprocessing (Excel)
- Exploratory Data Analysis using Python and Jupyter Notebook
- Revenue, product, country, and customer-level analysis
- Customer segmentation using DAX (Power BI)
- Interactive dashboard development with filters: Year, Country, Product, Customer Segment
- Business insight generation and reporting

Out of Scope

- Real-time or live data streaming integration
- Predictive or machine learning modelling
- Customer-level demographics or behavioural profiling beyond revenue
- Integration with external CRM or ERP systems

5. Feasibility Study

5.1 Technical Feasibility

All tools used in this project , Microsoft Excel, Python (pandas, matplotlib, seaborn), Jupyter Notebook, and Microsoft Power BI Desktop are industry-standard, widely adopted, and freely or commercially available. The dataset (Excel format) is directly importable into all three platforms. The hardware requirements are minimal, making this solution feasible on any modern personal computer.

5.2 Operational Feasibility

The resulting Power BI dashboard is designed for non-technical business users. Interactive slicers allow filtering by Year, Country, Product, and Customer Segment without any programming knowledge. All visuals are self-explanatory with clearly labelled axes and KPI cards. The dashboard can be shared via Power BI Service for organisation-wide access.

5.3 Economic Feasibility

The entire project utilises open-source or freely available tools. Python and Jupyter Notebook are free. Power BI Desktop is available at no cost. The dataset is publicly available from the UCI ML Repository. There are no licensing costs, making this project economically viable with zero direct expenditure.

6. Methodology / Implementation

This project follows a structured seven-stage data analytics pipeline:

6.1 Data Collection

Dataset: UCI Machine Learning Repository Online Retail Dataset

Source	UCI Machine Learning Repository
Format	Microsoft Excel (.xlsx)
Raw Rows	540,456 transactions
Columns	8 (InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country)
Time Period	December 2010 – December 2011
Geography	Primarily United Kingdom + 36 international markets

Raw Dataset :

F2 | 2.55

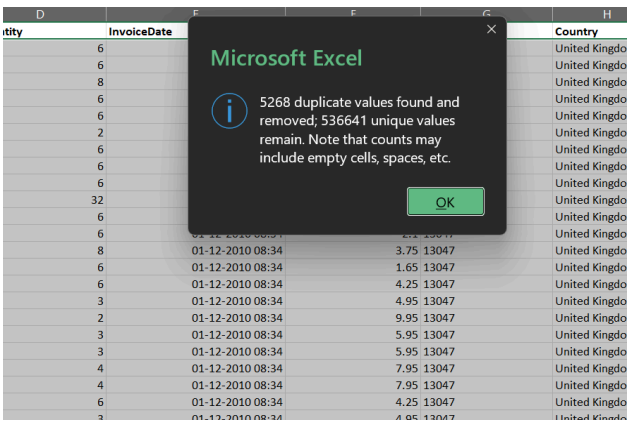
	A	B	C	D	E	F	G	H
1	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
2	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDE	6	01-12-2010 08:26	2.55	17850	United Kingdom
3	536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom
4	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom
5	536365	84029G	KNITTED UNION FLAG HOT WATER BOTT	6	01-12-2010 08:26	3.39	17850	United Kingdom
6	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850	United Kingdom
7	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	01-12-2010 08:26	7.65	17850	United Kingdom
8	536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-2010 08:26	4.25	17850	United Kingdom
9	536366	22633	HAND WARMER UNION JACK	6	01-12-2010 08:28	1.85	17850	United Kingdom
10	536366	22632	HAND WARMER RED POLKA DOT	6	01-12-2010 08:28	1.85	17850	United Kingdom
11	536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010 08:34	1.69	13047	United Kingdom
12	536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	01-12-2010 08:34	2.1	13047	United Kingdom
13	536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	01-12-2010 08:34	2.1	13047	United Kingdom
14	536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-2010 08:34	3.75	13047	United Kingdom
15	536367	22310	IVORY KNITTED MUG COSY	6	01-12-2010 08:34	1.65	13047	United Kingdom
16	536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOON	6	01-12-2010 08:34	4.25	13047	United Kingdom
17	536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-2010 08:34	4.95	13047	United Kingdom
18	536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	01-12-2010 08:34	9.95	13047	United Kingdom
19	536367	21754	HOME BUILDING BLOCK WORD	3	01-12-2010 08:34	5.95	13047	United Kingdom
20	536367	21755	LOVE BUILDING BLOCK WORD	3	01-12-2010 08:34	5.95	13047	United Kingdom
21	536367	21777	RECIPE BOX WITH METAL HEART	4	01-12-2010 08:34	7.95	13047	United Kingdom
22	536367	48187	DOORMAT NEW ENGLAND	4	01-12-2010 08:34	7.95	13047	United Kingdom
23	536368	22960	JAM MAKING SET WITH JARS	6	01-12-2010 08:34	4.25	13047	United Kingdom
24	536368	22913	RED COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047	United Kingdom

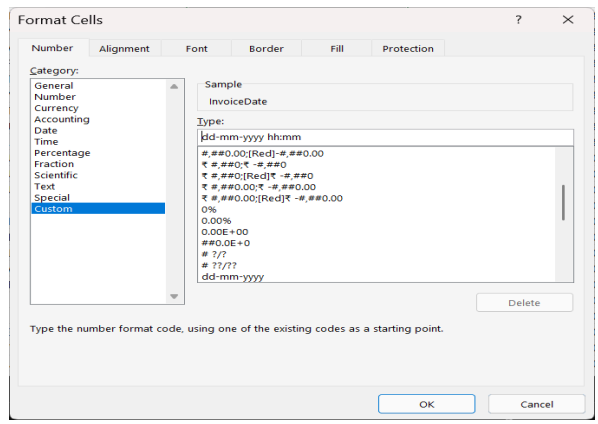
6.2 Data Cleaning and Transformation (Microsoft Excel)

Initial preprocessing was performed in Excel to prepare a clean dataset for downstream analysis:

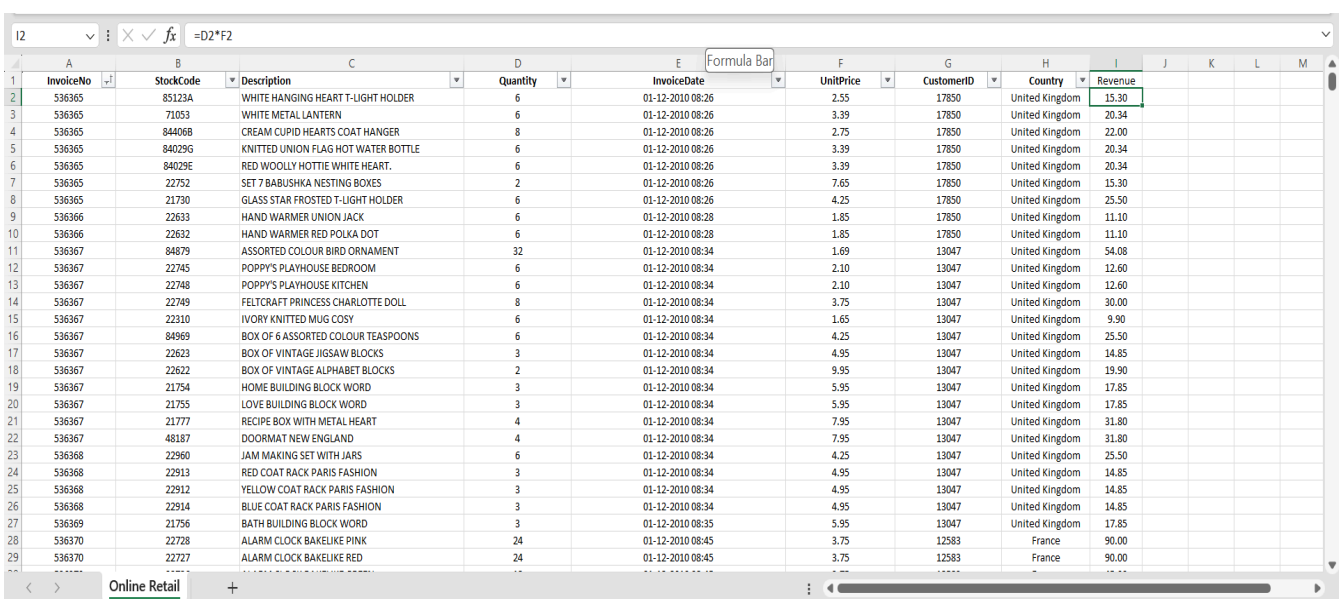
- Removed duplicate records using Remove Duplicates function
- Validated and corrected data types: Quantity (Integer), UnitPrice (Float),etc.
- Created a derived Revenue column: Revenue = Quantity × UnitPrice
- Removed transactions with zero UnitPrice (sample/error records)

Result: 401,605 clean, structured records ready for EDA and Power BI transformation..





Cleaned Dataset : [View Cleaned Dataset]



InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850	United Kingdom	15.30
536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850	United Kingdom	20.34
536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850	United Kingdom	22.00
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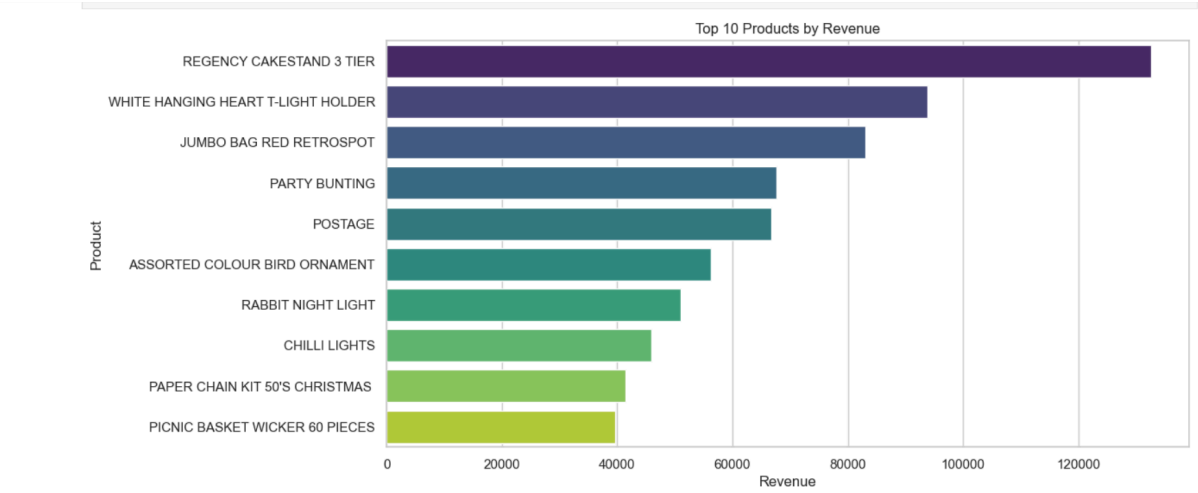
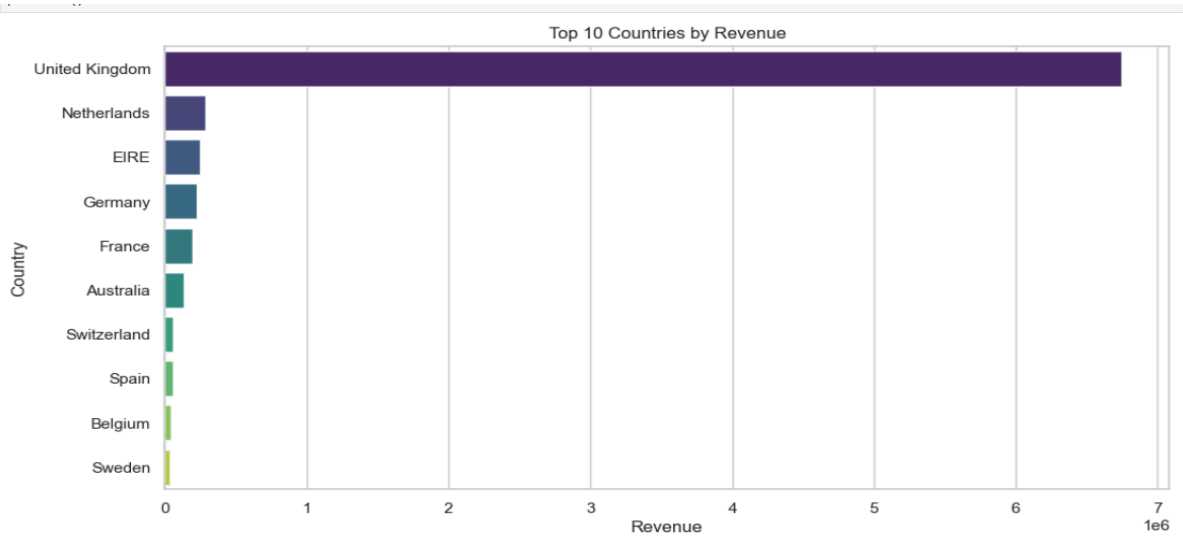
536365	21730	GLASS STAR FROSTED T-LIGHT HOLDER	6	01-12-2010 08:26	4.25	17850	United Kingdom	25.50
536366	22633	HAND WARMER UNION JACK	6	01-12-2010 08:28	1.85	17850	United Kingdom	11.10
536366	22632	HAND WARMER RED POLKA DOT	6	01-12-2010 08:28	1.85	17850	United Kingdom	11.10
536367	84879	ASSORTED COLOUR BIRD ORNAMENT	32	01-12-2010 08:34	1.69	13047	United Kingdom	54.08
536367	22745	POPPY'S PLAYHOUSE BEDROOM	6	01-12-2010 08:34	2.10	13047	United Kingdom	12.60
536367	22748	POPPY'S PLAYHOUSE KITCHEN	6	01-12-2010 08:34	2.10	13047	United Kingdom	12.60
536367	22749	FELTCRAFT PRINCESS CHARLOTTE DOLL	8	01-12-2010 08:34	3.75	13047	United Kingdom	30.00
536367	22310	IVORY KNITTED MUG COSY	6	01-12-2010 08:34	1.65	13047	United Kingdom	9.90
536367	84969	BOX OF 6 ASSORTED COLOUR TEASPOONS	6	01-12-2010 08:34	4.25	13047	United Kingdom	25.50
536367	22623	BOX OF VINTAGE JIGSAW BLOCKS	3	01-12-2010 08:34	4.95	13047	United Kingdom	14.85
536367	22622	BOX OF VINTAGE ALPHABET BLOCKS	2	01-12-2010 08:34	9.95	13047	United Kingdom	19.90
536367	21754	HOME BUILDING BLOCK WORD	3	01-12-2010 08:34	5.95	13047	United Kingdom	17.85
536367	21755	LOVE BUILDING BLOCK WORD	3	01-12-2010 08:34	5.95	13047	United Kingdom	17.85
536367	21777	RECIPE BOX WITH METAL HEART	4	01-12-2010 08:34	7.95	13047	United Kingdom	31.80
536367	48187	DOORMAT NEW ENGLAND	4	01-12-2010 08:34	7.95	13047	United Kingdom	31.80
536368	22960	JAM MAKING SET WITH JARS	6	01-12-2010 08:34	4.25	13047	United Kingdom	25.50
536368	22913	RED COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047	United Kingdom	14.85
536368	22912	YELLOW COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047	United Kingdom	14.85
536368	22914	BLUE COAT RACK PARIS FASHION	3	01-12-2010 08:34	4.95	13047	United Kingdom	14.85
536369	21756	BATH BUILDING BLOCK WORD	3	01-12-2010 08:35	5.95	13047	United Kingdom	17.85
536370	22728	ALARM CLOCK BAKELIKE PINK	24	01-12-2010 08:45	3.75	12583	France	90.00
536370	22727	ALARM CLOCK BAKELIKE RED	24	01-12-2010 08:45	3.75	12583	France	90.00

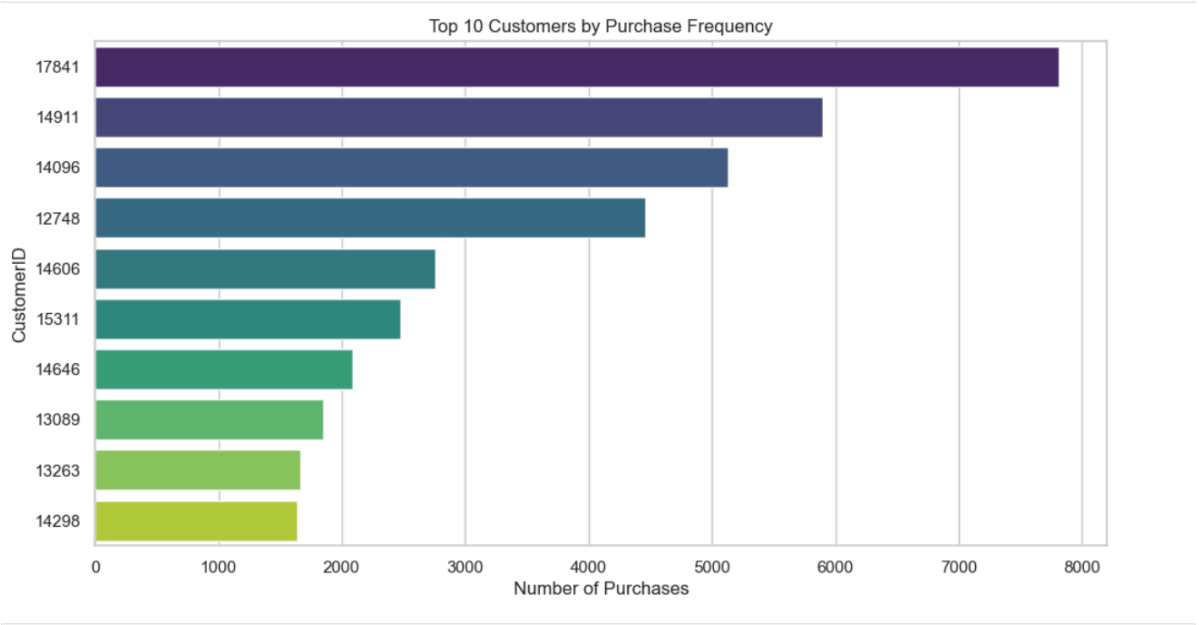
6.3 Exploratory Data Analysis (Python — Jupyter Notebook)

Libraries used: pandas, matplotlib, seaborn

- Loaded dataset: df = pd.read_excel(file_path)
- Inspected data: df.head(), df.info(), df.describe()
- Identified data anomalies: negative quantities, zero prices, missing CustomerIDs
- Grouped by Description to identify Top 10 Revenue-generating products
- Aggregated revenue by Country to identify geographic concentration
- Used value_counts() on CustomerID to find high-frequency purchasers
- Extracted YearMonth from InvoiceDate for monthly revenue trend analysis






Top 10 Customers by Purchase Frequency
CustomerID
17841
14911
14096
12748
14606
15311
14646
13089
13263
14298
0
1000
2000
3000
4000
5000
6000
7000
8000
Number of Purchases

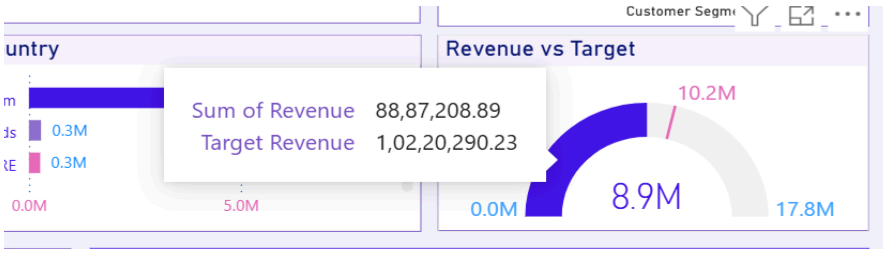
Monthly Revenue Trend
1e6
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Revenue
2010-12
2011-01
2011-02
2011-03
2011-04
2011-05
2011-06
2011-07
2011-08
2011-09
2011-10
2011-11
2011-12
Month

6.4 DAX Calculations (Power Bi)

Target Revenue

```
TargetRevenue = SUM('Online Retail'[Revenue]) * 1.15
```

This measure sets a 15% growth target above actual revenue, establishing a performance benchmark visible on the dashboard gauge.



Customer Segmentation

A DAX calculated column classifies each customer based on their total revenue contribution:

Segment	Revenue Threshold
Champions	Greater than £10,000
Loyal	£5,001 – £10,000
At Risk	£1,001 – £5,000
Lost	£1,000 or below

The CALCULATE and ALLEXCEPT functions aggregate revenue per CustomerID across all transactions, enabling accurate segment assignment.

```
CustomerSegment = VAR TotalRevenuePerCustomer =
 CALCULATE( SUM ('Online Retail'[Revenue] ),
        ALLEXCEPT ('Online Retail', 'Online Retail'[CustomerID] ) )
RETURN
IF( TotalRevenuePerCustomer > 10000, "Champions",
    IF(TotalRevenuePerCustomer > 5000, "Loyal",
        IF(TotalRevenuePerCustomer > 1000, "At Risk", "Lost") ) )
```

6.5 Dashboard Development (Power BI)

The interactive dashboard was designed with the following components:

- KPI Cards: Total Revenue, Target Revenue, Total Orders, Unique Customers, Unique Products, Countries
- Line Chart: Monthly revenue trend (Jan 2011 – Dec 2011)
- Bar Chart: Top products by revenue
- Horizontal Bar Chart: Revenue by country
- Pie Chart: Revenue by quarter
- Clustered Bar Chart: Customer segmentation by revenue
- Gauge Visual: Revenue vs Target (£8.9M actual vs £10.2M target)
- Data Table: Top customers with CustomerID, Country, Revenue, Quantity, Product
- Slicers: Year, Country, Product, Customer Segment

6.6 Data Pipeline Summary

The three-stage cleaning pipeline produced the following record counts:

Stage	Tool	Records	Notes
Raw Dataset	UCI Repository	540,456	Original download
After Excel Cleaning	Microsoft Excel	401,605	Duplicates removed, revenue derived
After Power BI Filtering	Power BI / DAX	392,692	Returns, zero-price records excluded

6.7 Testing & Validation

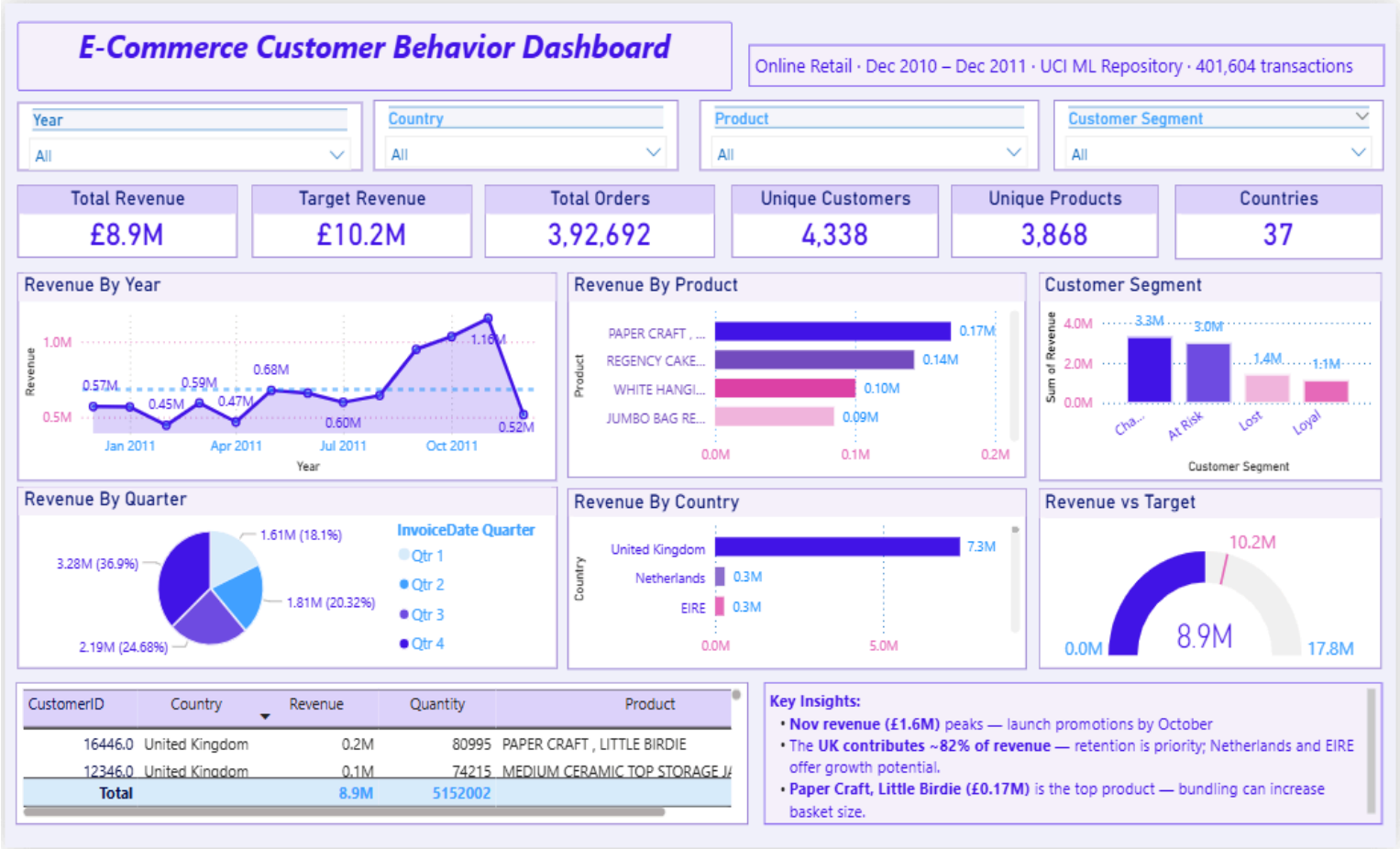
- Cross-validated KPI totals against Python EDA outputs to ensure consistency
- Verified revenue calculations (Quantity × UnitPrice) match across Excel, Python, and Power BI
- Tested all slicer combinations to confirm correct dynamic filtering
- Validated customer segment counts against raw group-by queries
- Confirmed total order count matches post-filter transaction count (392,692)

Power BI Table View :

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue	CustomerSegment
536557	22731	3D CHRISTMAS STAMPS STICKERS	1	01-12-2010 14:41:00	1.25	17841	United Kingdom	1.25	Champions
536557	22953	BIRTHDAY PARTY CORDON BARRIER TAPE	1	01-12-2010 14:41:00	1.25	17841	United Kingdom	1.25	Champions
536557	21672	WHITE SPOT RED CERAMIC DRAWER KNOB	1	01-12-2010 14:41:00	1.25	17841	United Kingdom	1.25	Champions
536557	22686	FRENCH BLUE METAL DOOR SIGN No	1	01-12-2010 14:41:00	1.25	17841	United Kingdom	1.25	Champions
536557	20963	APPLE BATH SPONGE	1	01-12-2010 14:41:00	1.25	17841	United Kingdom	1.25	Champions
536984	21446	12 RED ROSE PEG PLACE SETTINGS	1	03-12-2010 14:35:00	1.25	17841	United Kingdom	1.25	Champions
536984	20963	APPLE BATH SPONGE	1	03-12-2010 14:35:00	1.25	17841	United Kingdom	1.25	Champions
536984	20961	STRAWBERRY BATH SPONGE	1	03-12-2010 14:35:00	1.25	17841	United Kingdom	1.25	Champions
537405	22673	FRENCH GARDEN SIGN BLUE METAL	1	06-12-2010 14:51:00	1.25	17841	United Kingdom	1.25	Champions
537405	20972	PINK CREAM FELT CRAFT TRINKET BOX	1	06-12-2010 14:51:00	1.25	17841	United Kingdom	1.25	Champions
538163	22679	FRENCH BLUE METAL DOOR SIGN 4	1	09-12-2010 17:27:00	1.25	17841	United Kingdom	1.25	Champions
538163	20971	PINK BLUE FELT CRAFT TRINKET BOX	1	09-12-2010 17:27:00	1.25	17841	United Kingdom	1.25	Champions
538163	21867	PINK UNION JACK LUGGAGE TAG	1	09-12-2010 17:27:00	1.25	17841	United Kingdom	1.25	Champions
538866	22757	LARGE RED BABUSHKA NOTEBOOK	1	14-12-2010 14:40:00	1.25	17841	United Kingdom	1.25	Champions
539469	21447	12 IVORY ROSE PEG PLACE SETTINGS	1	19-12-2010 13:59:00	1.25	17841	United Kingdom	1.25	Champions
539469	84754	S/15 SILVER GLASS BAUBLES IN BAG	1	19-12-2010 13:59:00	1.25	17841	United Kingdom	1.25	Champions
540021	20717	STRAWBERRY SHOPPER BAG	1	04-01-2011 12:51:00	1.25	17841	United Kingdom	1.25	Champions
540021	20718	RED RETROSPOT SHOPPER BAG	1	04-01-2011 12:51:00	1.25	17841	United Kingdom	1.25	Champions
540021	20716	PARTY FOOD SHOPPER BAG	1	04-01-2011 12:51:00	1.25	17841	United Kingdom	1.25	Champions

Table: Online Retail (4,01,604 rows) Column: Revenue (4,041 distinct values)

Power BI Dashboard :



7. Tools and Technologies Used

Tool / Technology	Version / Type	Purpose
Microsoft Excel	2016+	Data ingestion, cleaning, Revenue column creation
Python	3.x	Exploratory Data Analysis (EDA)
pandas	Library	Data manipulation and aggregation
matplotlib	Library	Data visualisation (EDA charts)
seaborn	Library	Statistical visualisation
Jupyter Notebook	IDE	Interactive Python analysis environment
Microsoft Power BI	Desktop	Data transformation, DAX, dashboard development
DAX	Formula Language	Custom measures and customer segmentation
UCI ML Repository	Dataset Source	Online Retail Dataset (public domain)

8. Results and Output

8.1 Key Performance Indicators

Metric	Value	Benchmark
Total Revenue	£8.9 Million	Target: £10.2M
Total Orders	392,692	Post-cleaning
Unique Customers	4,338	Active buyers
Unique Products	3,868	SKUs sold
Countries Served	37	Global reach
Revenue Gap vs Target	£1.3 Million	12.7% shortfall vs target

8.2 Revenue Trend Insights

Revenue showed a consistent upward trend throughout the year, increasing from £0.45M in January 2011 to a peak of £1.16M in November 2011, indicating strong business growth over time.

A significant spike in Q4 (October–December) highlights pronounced seasonal demand, most likely driven by Christmas and festive purchasing behavior.

November 2011 emerges as the highest-performing month, suggesting that customers tend to complete the majority of their holiday purchases before December.

December 2011 records a decline to £0.52M, which indicates early order fulfilment and reduced post-peak purchasing activity.

Overall, Q4 contributes approximately 36.9% (£3.28M) of total annual revenue, making it the most critical quarter for business performance.

8.3 Geographic Analysis

The United Kingdom dominates revenue generation, contributing approximately £7.3M (~82%) of total revenue, indicating a highly concentrated primary market.

Netherlands (£0.3M) and EIRE (£0.3M) are the second and third largest contributors, representing key international markets with growth potential.

The remaining 36 countries collectively contribute around 18% of total revenue, reflecting a broad but less penetrated global presence.

8.4 Product Performance

The top-performing product, Paper Craft, Little Birdie, generated approximately £0.17M in revenue, making it the highest revenue contributor.

The second-best product, Regency Cakestand 3-Tier, generated around £0.14M, showing consistent demand among customers.

The top 10 products are primarily from home décor, gifting, and craft categories, indicating strong customer preference for these segments.

Despite having a large catalog of 3,868 unique products (SKUs), revenue is highly concentrated among a relatively small number of top-performing items.

This highlights the importance of focusing on high-demand products for promotions, bundling, and inventory optimization.

8.5 Customer Segmentation

Champions (> £10,000) represent the highest-value customers, contributing significantly to total revenue and requiring priority retention strategies and exclusive engagement.

Loyal Customers (£5,001–£10,000) form a strong and stable revenue base, making them ideal targets for upselling and cross-selling strategies.

At Risk Customers (£1,001–£5,000) represent a large portion of the customer base and require targeted re-engagement efforts to prevent churn.

Lost Customers (< £1,000) contribute minimally to revenue and can be targeted through win-back campaigns or promotional incentives.

This segmentation enables more personalized marketing strategies, improving customer retention, lifetime value, and overall revenue growth.

9. Challenges Faced and Solutions

Challenge	Solution Applied
Negative quantity and revenue values from returns/cancellations	Applied filters: Revenue > 0, Quantity > 0, UnitPrice > 0 in Power BI Power Query
Missing CustomerIDs for guest transactions	Retained transactions with missing CustomerIDs for revenue analysis; excluded from customer-count KPIs
Large dataset (540K rows) causing slow processing	Filtered dataset to 392K valid records; used Power BI's columnar storage for performance
Customer segmentation requiring cross-customer aggregation in DAX	Used CALCULATE with ALLEXCEPT to compute per-customer totals across the full dataset
Revenue target definition without historical benchmark	Set target at 15% above actual revenue as a standard growth benchmark using DAX measure

Ensured data consistency by standardizing formats and validating key fields before analysis, improving overall data reliability.

Optimized performance by reducing unnecessary columns and focusing only on relevant variables for analysis and visualization.

Maintained a balance between data accuracy and completeness by carefully handling missing values without losing critical revenue information.

\10. Advantages

- End-to-end pipeline from raw data to actionable insights in a single, cohesive workflow
- Interactive filtering enables stakeholders to explore data without technical knowledge
- Customer segmentation directly informs personalised marketing campaigns
- Seasonal trend analysis enables proactive, data-driven campaign scheduling
- Revenue vs Target gauge provides instant performance visibility for management
- Scalable architecture the Power BI model can be updated with new data without redesign
- Zero licensing cost all tools are free or widely available within corporate environments

11. Limitations

- Dataset is historical (2010–2011) and does not reflect current market conditions
- Customer segmentation is revenue-only; it does not incorporate recency or frequency dimensions of full RFM analysis
- Missing CustomerIDs (~25% of transactions) limit completeness of customer-level insights
- The dashboard does not support real-time or scheduled data refresh without Power BI Service and a data gateway
- Geographic analysis is limited to country level regional or city-level insights are unavailable in this dataset
- No predictive capability the dashboard is purely descriptive and diagnostic, not prescriptive

12. Future Enhancements

- Implement full RFM (Recency-Frequency-Monetary) scoring for more sophisticated customer segmentation
- Integrate predictive analytics using Python's scikit-learn for demand forecasting and churn prediction
- Connect Power BI to a live SQL database or cloud data warehouse (Azure Synapse, Snowflake) for real-time refresh
- Add basket/market basket analysis (Apriori algorithm) to identify product bundling opportunities
- Incorporate customer lifetime value (CLV) modelling to prioritise acquisition and retention spend
- Deploy the dashboard via Power BI Service for mobile access and automated scheduled refresh
- Develop a recommendation engine to personalise product suggestions based on purchase history

13. Conclusion

This project successfully demonstrates a complete, professional-grade data analytics pipeline applied to real-world e-commerce transactional data. Starting from a raw dataset of over 540,000 records, the project systematically cleaned, explored, transformed, and visualised the data to produce an interactive Power BI dashboard that delivers genuine business value.

The dashboard reveals that the business generated £8.9M in revenue across 37 countries from 4,338 unique customers, with the United Kingdom accounting for approximately 82% of total revenue. A clear November revenue peak confirms strong seasonal demand, offering a strategic window for targeted promotions. Customer segmentation highlights distinct groups that warrant differentiated marketing approaches from VIP retention for Champions to re-engagement campaigns for the At Risk segment.

The revenue gap of £1.3M against the £10.2M target presents a clear and quantified growth opportunity. The product analysis identifies bundling candidates, while the geographic data points to Netherlands and EIRE as priority international expansion markets.

This project confirms that a structured data analytics approach , even with freely available tools can transform raw transactional records into strategic, decision-ready insights that directly support revenue growth, customer retention, and market expansion objectives.

14. References

- UCI Machine Learning Repository: Online Retail Dataset. https://archive.ics.uci.edu/dataset/352/online+retail
- Chen, D., Sain, S. L., & Guo, K. (2012). Data mining for the online retail industry: A case study of RFM model-based customer segmentation using data mining. Journal of Database Marketing & Customer Strategy Management, 19(3), 197–208.
- Microsoft. (2025). Power BI Documentation. https://learn.microsoft.com/en-us/power-bi/
- McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media.
- Tools used: Python, Power BI Desktop, Microsoft Excel

15. Appendix

Appendix A: Dashboard Overview

The Power BI dashboard (E-Commerce Customer Behaviour Dashboard) includes the following components as visible in the project screenshot:

- KPI Cards Row: Total Revenue (£8.9M), Target Revenue (£10.2M), Total Orders (392,692), Unique Customers (4,338), Unique Products (3,868), Countries (37)
- Revenue By Year (Monthly Trend): Line chart showing monthly revenue from Jan 2011 to Dec 2011, with November 2011 as the peak month (£1.16M)
- Revenue By Product: Horizontal bar chart — top 4 products by revenue
- Customer Segment: Clustered bar chart — Champions, At Risk, Lost, Loyal by revenue
- Revenue By Quarter: Pie chart — Q1 18.1%, Q2 20.32%, Q3 24.68%, Q4 36.9%
- Revenue By Country: Horizontal bar chart — UK (£7.3M), Netherlands (£0.3M), EIRE (£0.3M)
- Revenue vs Target: Gauge visual — £8.9M actual vs £10.2M target
- Customer Detail Table: Top customers with CustomerID, Country, Revenue, Quantity, Product
- Key Insights Panel: Summarised business recommendations

Appendix B: Key DAX Formulas

Target Revenue:

```
TargetRevenue = SUM('Online Retail'[Revenue]) * 1.15
```

Customer Segmentation Column:

```
CustomerSegment = VAR TotalRev = CALCULATE(SUM([Revenue]), ALLEXCEPT('Online Retail',
[CustomerID])) RETURN IF(TotalRev > 10000, "Champions", IF(TotalRev > 5000, "Loyal",
IF(TotalRev > 1000, "At Risk", "Lost")))
```

Appendix C: Project Files

- Power BI Dashboard (.pbix)
- Python Notebook (.ipynb)
- Screenshots (.png)
- Dataset: View Cleaned Dataset
- Project Report (.pdf)
- README.md